Chapter 11: Hashing and Collision

Technical / interview questions, exercises, multiple choice questions and answer key for Chapter 11 (Hashing and Collision) of Data Structures.

Programming Exercises

1. Write a program to implement hashing using the Division Method.
2. Implement a Hash Table with user-defined table size.
3. Write a program to calculate load factor after each insertion.
4. Implement separate chaining using linked lists.
5. Implement open addressing using linear probing.
6. Implement quadratic probing with suitable constants.
7. Implement double hashing using two hash functions.
8. Compare collision counts for linear and quadratic probing.
9. Write a program to detect primary clustering.
10. Write a program to detect secondary clustering.
11. Implement deletion in open addressing using tombstones.
12. Measure average search time for different load factors.
13. Implement dynamic resizing (rehashing).
14. Write a program to store strings using hashing.
15. Implement folding method for long numeric keys.
16. Implement mid-square hashing.
17. Convert keys from base 13 to base 10 and hash them.
18. Simulate universal hashing.
19. Create a Hash Table for storing student roll numbers.
20. Design a Hash Table for storing PIN codes.
21. Compare memory usage of chaining and open addressing.
22. Insert 1000 random numbers and calculate collision rate.
23. Write a menu-driven Hash Table program.
24. Implement Hash Table using arrays only (no STL).
25. Write a program to search and count probe length.
26. Implement Hash Table for storing employee IDs.
27. Write a function to calculate clustering length.
28. Implement hashing for alphanumeric keys.
29. Write a performance comparison report.
30. Simulate worst-case collision scenario.
31. Implement hash map using object-oriented programming.
32. Write a program to measure time complexity experimentally.

Technical / Interview Questions

1. What is hashing and why is it used?
2. What is the difference between hashing and indexing?
3. Define load factor.
4. Why should table size be prime?

5. What is a collision?
6. What are collision resolution techniques?
7. Difference between chaining and open addressing?
8. What is primary clustering?
9. What is secondary clustering?
10. Why is double hashing better than linear probing?
11. What is universal hashing?
12. Explain worst-case time complexity of hashing.
13. What happens when load factor exceeds 1?
14. Why is deletion difficult in open addressing?
15. What is rehashing?
16. When should resizing be done?
17. What is a tombstone?
18. Compare Hash Table with binary search tree.
19. Can hashing be used for ordered data?
20. How is hashing used in compilers?
21. How are passwords stored using hashing?
22. What is the difference between hash map and Hash Table?
23. What makes a good hash function?
24. Explain multiplication method formula.
25. Why does linear probing cause clustering?
26. Explain double hashing formula.
27. What is average time complexity of search?
28. What is worst-case complexity?
29. Can two different keys have same hash value?
30. What is dynamic hashing?
31. How is hashing used in databases?
32. When should hashing not be used?

Multiple Choice Questions

1. The average time complexity of search in hashing is:

(a) O(n)
(b) O(log n)
(c) O(1)
(d) $O(n^2)$

2. Load factor (α) is defined as:

(a) m/n
(b) n/m
(c) n^2/m
(d) m^2/n

3. A collision occurs when:

(a) Two keys are identical
(b) Two keys map to the same index
(c) Table is full
(d) Load factor is zero

4. For the division method, the table size should preferably be:

(a) Even number
(b) Power of 2
(c) Prime number
(d) Odd number

5. Linear probing suffers from:

(a) Secondary clustering
(b) Primary clustering
(c) No clustering
(d) Rehashing

6. Quadratic probing helps to reduce:

(a) Primary clustering
(b) Secondary clustering
(c) Overflow
(d) Load factor

7. Double hashing uses:

(a) One hash function
(b) Two hash functions
(c) Three hash functions
(d) No hash function

8. In separate chaining, each table index stores:

(a) Stack
(b) Queue
(c) Linked list
(d) Tree always

9. Worst-case time complexity of hashing search is:

(a) O(1)
(b) O(log n)
(c) O(n)
(d) O(n log n)

10. Universal hashing aims to reduce:

(a) Memory usage
(b) Execution time
(c) Collision probability
(d) Table size

11. Folding method is most suitable for:

(a) Short keys
(b) Long numeric keys
(c) Boolean values
(d) Single digit keys

12. Radix conversion hashing involves:

(a) Modulo operation
(b) Base conversion
(c) Random selection
(d) Bit shifting only

13. Deletion in open addressing requires:

(a) Clearing cell completely
(b) Tombstone marker
(c) Linked list
(d) Tree balancing

14. A good hash function should be:

(a) Slow
(b) Deterministic and uniform
(c) Complex only
(d) Dependent on input order

15. α represents:

(a) Table size
(b) Number of keys
(c) Load factor
(d) Collision rate

16. Primary clustering occurs in:

(a) Double hashing
(b) Quadratic probing
(c) Linear probing
(d) Chaining

17. Secondary clustering occurs in:

(a) Linear probing
(b) Quadratic probing
(c) Chaining
(d) Division method

18. Rehashing is performed to:

(a) Increase collisions
(b) Reduce load factor
(c) Delete all keys
(d) Reduce memory

19. Hash Tables are best suited for:

(a) Ordered traversal
(b) Sorting
(c) Fast search
(d) Graph traversal

20. In double hashing, the second hash function must be:

(a) Zero
(b) Constant
(c) Non-zero
(d) Negative

21. In chaining, load factor can be:

(a) Less than 0
(b) Equal to 0 only
(c) Greater than 1
(d) Always less than 1

22. In open addressing, elements are stored:

(a) Outside table
(b) In separate memory
(c) Inside the table
(d) In files

23. The formula h(k) = k mod m belongs to:

(a) Folding method
(b) Multiplication method
(c) Division method
(d) Mid-square method

24. Collisions are unavoidable because:

(a) Keys are infinite
(b) Table size is finite
(c) Hash functions are weak
(d) Load factor is zero

25. Separate chaining uses:

(a) Arrays only
(b) Linked lists
(c) Stacks
(d) Recursion

26. Linear probing probe sequence is:

(a) h(k) + i
(b) h(k) × i
(c) h(k) − i
(d) $h(k)^2$

27. Double hashing reduces:

(a) Clustering
(b) Table size
(c) Keys
(d) Memory

28. Hashing is not suitable for:

(a) Fast lookup
(b) Indexing
(c) Ordered data traversal
(d) Password storage

29. Tombstone markers are used in:

(a) Chaining
(b) Linear search
(c) Open addressing
(d) Sorting

30. A hash function must always be:

(a) Random every time
(b) Deterministic
(c) Recursive
(d) Exponential

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	2	B	3	B	4	C	5	B
6	A	7	B	8	C	9	C	10	C
11	B	12	B	13	B	14	B	15	C
16	C	17	B	18	B	19	C	20	C
21	C	22	C	23	C	24	B	25	B
26	A	27	A	28	C	29	C	30	B